#HW 1

Graded

Student

Samantha Rangel

Total Points

70 / 100 pts

Question 1

Q1

18 / 20 pts

– 2 pts There is an error in your calculation. Feet should be converted to inches. ft^3 can be converted to inch^3 by multiplying it with 12^3.

– 0 pts Click here to replace this description.

✔ **– 2 pts** Since the Patm is given in lbf so you should get your gauge and system pressure in lbf too. So you should divide rho*g*h by the gravity.

– 2 pts Absolute or system pressure can be obtained by adding up the gauge pressure with Patm.

– 2 pts The effective height should be 15*sin 40 degrees. You would have got a different answer if you had used 40 degree instead of 30 degree since the question mentioned a 40 degree inclination.

– 5 pts Late

Question 2

Q2

15 / 20 pts

– 0 pts Click here to replace this description.

– 4 pts Since, the specific volume has been given, in order to get the density, you need to take the reciprocal of the specific volume.

– 1 pt The answer you got is in Pascal. To convert it to kPa you must divide it by 1000.

✔ **– 5 pts** Incomplete submission

– 5 pts Late

Question 3

Q3

10 / 20 pts

– 0 pts Click here to replace this description.

– 2 pts In the question it has been mentioned that the pressure changes from 1 bar to 200 bar in increments and volume should be calculated next. You had increased volume and then calculated pressure. You are supposed to increase the pressure.

– 2 pts Pressure should be on the y axis and volume should be on the x axis

– 2 pts Please remember to share your data from the next time. The question also mentioned that you should share the screenshot of both the plot and data.

– 2 pts The question said to add 20 increments. So you should have reported 20 data sets.

– 2 pts Please remember to add titles on your axes from the next time.

✔ **– 10 pts** Incomplete

– 5 pts Late

– 1 pt Add units on axes.

Question 4

Q4

12 / 15 pts

– 0 pts Click here to replace this description.

✔ **– 2 pts** The temperature change is not the same for all. It is the same for C and K. It is the same for F and R. You should clearly state it.

– 2 pts The conversion scale for F to C is wrong. It is (F-32)/1.8 i.e. the entire (F-32) gets multiplied by 5/9 or divided by 9/5.

– 5 pts Incomplete

✔ **– 1 pt** You should show the conversion of both 70 degree F and 42 degree F. You have not shown conversion for the 70 degree F.

– 5 pts Late

Question 5

Q5

15 / 25 pts

– 0 pts Click here to replace this description.

– 2 pts You are supposed to calculate the absolute pressure in 5a. So you add the Patm too for getting that.

– 1 pt Pressure inside the submersible is same as the atmospheric pressure.

– 1 pt You should deduct Patm from Pabs to get gauge pressure

✔ **– 10 pts** Incomplete

– 10 pts Late

Question assigned to the following page: 1

#1

a)

$$P_{absolute} = P_{atm} + (h \cdot \rho \cdot g)$$

$$= 14.7 + (9.642 \cdot 0.0314 \cdot 32.2)$$

$$= 24.45 \ lbf/in^3$$

$$\sin 40 = \frac{h}{15in}$$

$$h = 15\sin(40)$$

$$= 9.642$$

$$\rho = \frac{54.2 \ lb}{ft^3} = \frac{54.2lb}{(12in)^3} = 0.0314 \ lb/in^3$$

b)

$$P_{gage} = P_{absolute} - P_{atm}$$

$$= 24.45 - 14.7$$

$$= 9.75 \ lbf/in^3$$

c) liquids moves with less difficulty on a linclined Plane compared to a U shaped Plane.

#4

a. $T = 42°F = 5.56°c = 278.56K = 501.408 R$

b. $\Delta T = 70°F - 42°F = 28°F = -2.22°C = 270.78K = 487.404 R$

c. The Temperature change from °F to °R is 1.8 times difference

e. The temperature Change from °K to °C is it goes from a larger to smaller scale.

#2

$$\rho = \frac{1}{v}$$

$$h = 0.12$$

$$P_{a_1} - P_{a_2} = \rho \cdot g(H_1 - H_2) + P_{atm}$$

$$= \left(\frac{1}{0.00122}\right)(9.81)(12-0) + 101 \ kPa$$

Question assigned to the following page: 2

#1

a)

$P_{absolute} = P_{atm} + (h \cdot \rho \cdot g)$

$= 14.7 + (9.642 \cdot 0.0314 \cdot 32.2)$

$= 24.45\ lbf/in^3$

$\sin 40 = \frac{h}{15in}$

$h = 15\sin(40)$

$= 9.642$

$\rho = \frac{54.2\ lb}{ft^3} = \frac{54.2 lb}{(12in)^3} = 0.0314\ lb/in^3$

b)

$P_{gage} = P_{absolute} - P_{atm}$

$= 24.45 - 14.7$

$= 9.75\ lbf/in^3$

c) liquids moves with less difficulty on a linclined plane compared to a U shaped plane.

#4

a. $T = 42°F = 5.56°C = 278.56K = 501.408R$

b. $\Delta T = 70°F - 42°F = 28°F = -2.22°C = 270.78K = 487.404R$

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#2

$\rho = \frac{1}{v}$

$h = 0.12$

$P_{a_1} - P_{a_2} = \rho \cdot g (H_1 - H_2) + P_{atm}$

$= (\frac{1}{0.00122})(9.81)(12-0) + 101\ kPa$

Question assigned to the following page: 3

#1

a)

$P_{absolute} = P_{atm} + (h \cdot \rho \cdot g)$

$= 14.7 + (9.642 \cdot 0.0314 \cdot 32.2)$

$= 24.45 \ lbf/in^3$

$\sin 40 = \frac{h}{15in}$

$h = 15\sin(40)$

$= 9.642$

$\rho = \frac{54.2 \ lb}{ft^3} = \frac{54.2 lb}{(12in)^3} = 0.0314 \ lb/in^3$

b)

$P_{gage} = P_{absolute} - P_{atm}$

$= 24.45 - 14.7$

$= 9.75 \ lbf/in^3$

c) liquids moves with less difficulty on a linclined Plane compared to a U shaped Plane.

#4

a. $T = 42°F = 5.56°c = 278.56K = 501.408R$

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Question assigned to the following page: 4

#1

a)

$P_{absolute} = P_{atm} + (h \cdot \rho \cdot g)$

$= 14.7 + (9.642 \cdot 0.0314 \cdot 32.2)$

$= 24.45\ lbf/in^3$

$\sin 40 = \frac{h}{15in}$

$h = 15\sin(40)$

$= 9.642$

$\rho = \frac{54.2\ lb}{ft^3} = \frac{54.2 lb}{(12in)^3} = 0.0314\ lb/in^3$

b)

$P_{gage} = P_{absolute} - P_{atm}$

$= 24.45 - 14.7$

$= 9.75\ lbf/in^3$

c) liquids moves with less difficulty on a linclined Plane compared to a U shaped Plane.

#4

a. $T = 42°F = 5.56°C = 278.56K = 501.408R$

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$P_{a_1} - P_{a_2} = \rho \cdot g (H_1 - H_2) + P_{atm}$

$= (\frac{1}{0.00122})(9.81)(12-0) + 101\ kPa$

Question assigned to the following page: 5

#1

a)

$P_{absolute} = P_{atm} + (h \cdot \rho \cdot g)$

$= 14.7 + (9.642 \cdot 0.0314 \cdot 32.2)$

$= 24.45\ lbf/in^3$

$\sin 40 = \frac{h}{15in}$

$h = 15\sin(40)$

$= 9.642$

$\rho = \frac{54.2\ lb}{ft^3} = \frac{54.2lb}{(12in)^3} = 0.0314\ lb/in^3$

b)

$P_{gage} = P_{absolute} - P_{atm}$

$= 24.45 - 14.7$

$= 9.75\ lbf/in^3$

c) liquids moves with less difficulty on a linclined Plane compared to a U shaped Plane.

#4

a. $T = 42°F = 5.56°c = 278.56K = 501.408R$

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